

CARLOS MIGUEL RODRIGUEZ SANTOS

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer with 6 years of experience designing and deploying scalable ML solutions in production environments. Specialized in deep learning, computer vision, and NLP applications with a track record of reducing inference latency by up to 60% and improving model accuracy across multiple domains. Passionate about solving complex problems through data-driven approaches and mentoring junior engineers in machine learning best practices.

WORK EXPERIENCE

Senior Machine Learning Engineer

Telefonica Tech, Madrid, Spain

www.telefonica.com

November 2021 - Present

Designed and implemented a recommendation system serving 2.5 million daily active users, achieving 35% increase in click-through rates using collaborative filtering and deep neural networks.

Led a team of 4 ML engineers in developing computer vision models for automatic network infrastructure inspection, reducing on-site technician visits by 40%.

Optimized model inference pipelines using TensorFlow Lite and ONNX, reducing latency from 800ms to 320ms on edge devices deployed across the network.

Established MLOps best practices including automated retraining pipelines, A/B testing frameworks, and monitoring dashboards using Airflow and Prometheus.

Machine Learning Engineer

CloudPeak Technologies Ltd, Madrid, Spain

www.cloudpeak.io

January 2021 - October 2021

Developed end-to-end NLP pipeline for sentiment analysis of customer support tickets, processing 50,000+ documents daily with 91% accuracy using transformer-based models.

Implemented feature engineering workflows that reduced model training time by 45% through dimensionality reduction and intelligent caching strategies.

Collaborated with data engineering team to build ETL processes handling petabyte-scale datasets using Apache Spark and Delta Lake.

Presented quarterly stakeholder reports on model performance metrics and provided actionable insights driving product roadmap decisions.

Machine Learning Engineer

Indra Sistemas, Madrid, Spain

www.indracompany.com

March 2019 - March 2021

Built and deployed multiple time series forecasting models for financial transaction prediction, achieving RMSE improvements of 28% compared to baseline approaches.

Conducted extensive hyperparameter tuning and cross-validation experiments using scikit-learn and XGBoost, documenting results in reproducible experiment tracking systems.

Contributed to open-source computer vision library with contributions in object detection optimization, receiving 150+ GitHub stars.

Mentored 3 junior data scientists on machine learning fundamentals, statistical analysis, and best practices in model evaluation and validation.

EDUCATION

Master of Science in Machine Learning and Data Science

Universidad Politecnica de Madrid, Madrid, Spain

2017 - 2019

Bachelor of Science in Computer Science

Universidad Autonoma de Madrid, Madrid, Spain

2013 - 2017

TECHNICAL SKILLS

Programming Languages: Python, Java, Scala, SQL, R, Bash

Machine Learning Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Keras

Data Processing: Apache Spark, Pandas, NumPy, Polars, Dask

Data Warehousing: BigQuery, Snowflake, PostgreSQL, MongoDB

MLOps Tools: Airflow, MLflow, Kubeflow, Jenkins, Docker, Kubernetes

Cloud Platforms: Google Cloud Platform, AWS SageMaker, Microsoft Azure

Deep Learning Specializations: Computer Vision, Natural Language Processing, Time Series Analysis, Recommendation Systems

Version Control: Git, GitHub, GitLab

GITHUB REPOSITORIES

github.com/carlosmrodriguez/transformer-sentiment-classifier

Transformer-based sentiment analysis model with multilingual support, achieving 92% accuracy on Spanish financial documents.

github.com/carlosmrodriguez/timeseries-forecasting-toolkit

Comprehensive toolkit for time series forecasting including ARIMA, Prophet, and neural network implementations with hyperparameter optimization.

github.com/carlosmrodriguez/edge-inference-optimizer

Library for optimizing deep learning models for edge deployment, featuring quantization and pruning techniques reducing model size by 75%.

github.com/carlosmrodriguez/cv-annotation-pipeline

Automated computer vision annotation and labeling pipeline using semi-supervised learning approaches for reducing manual annotation effort.

CERTIFICATIONS AND AWARDS

Machine Learning Engineering for Production (MLOps) Certificate, DeepLearning.AI, 2022

TensorFlow Developer Certification, Google, 2021

Innovation Award for Recommendation System Project, Telefonica Tech, 2022

LANGUAGES

Spanish: Native

English: Fluent

Catalan: Professional Working Proficiency